



राष्ट्रीय प्रौद्योगिकी संस्थान दिल्ली

NATIONAL INSTITUTE OF TECHNOLOGY
DELHI

PROJECT REPORT

**TOPIC – PI Controlled Closed Loop Buck
Converter**

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ABSTRACT

The increasing demand for efficient and reliable power conversion systems in modern electronic devices has led to widespread adoption of DC-DC converters. Among them, the buck converter is a widely used step-down converter due to its simplicity, high efficiency, and ability to maintain a stable output voltage. However, conventional open-loop buck converters are unable to maintain a constant output voltage under varying input or load conditions. To address this limitation, this project presents the design and simulation of a closed-loop buck converter regulated by a Proportional-Integral (PI) controller.

The proposed system is modelled and analysed using MATLAB Simulink, with parameters such as input voltage of 12 V, desired output voltage of 5 V, switching frequency of 25 kHz, load resistance of $10\ \Omega$, inductance of $100\ \mu\text{H}$, and capacitance of $1000\ \mu\text{F}$. The PI controller dynamically adjusts the duty cycle of the Pulse Width Modulated (PWM) signal based on the error between the actual and reference output voltage, thereby ensuring voltage regulation and system stability.

Through simulation, the performance of the closed-loop system is evaluated under steady-state and transient conditions. The results indicate significant improvements in output voltage regulation, transient response, and reduced steady-state error compared to the open-loop counterpart. The simulation also confirms the controller's ability to effectively reject disturbances due to load changes. This project demonstrates the importance of control strategies in power electronics and provides a strong base for future enhancements such as hardware implementation, adaptive control methods, or integration with renewable energy sources.

INTRODUCTION

A DC-DC converter is an electronic device that directly converts one level of DC voltage to another. Among various types, the buck converter is one of the most widely used topologies, primarily employed to step down the input voltage to a lower level. It is commonly used in applications such as electric vehicles, trolley cars, forklifts, marine equipment, and industrial motor drives due to its ability to provide smooth acceleration, high efficiency, and fast dynamic response.

With the growing integration of renewable energy sources and the development of DC microgrid networks and smart home systems, the demand for high-quality, reliable, and efficient DC voltage conversion has increased. In such scenarios, buck converters play a vital role in maintaining stable power delivery.

To regulate the output voltage of a buck converter under varying input or load conditions, a closed-loop control system is essential. This study focuses on the design of a Proportional-Integral (PI) controller for a buck converter. The PI controller is implemented to ensure that the output voltage remains constant and stable despite fluctuations in the input supply or load demand. The objective of this project is to model and simulate a PI-controlled buck converter that delivers a fast, accurate, and reliable output response for modern power electronic applications..

OBJECTIVE

The main objective of this project is to design, model, and analyse a buck converter system regulated by a PI (Proportional-Integral) controller using MATLAB Simulink. The project aims to demonstrate how closed-loop control using feedback significantly enhances system performance. The key goals of this project are outlined as follows:

1. To Design and Simulate a Buck Converter

The first objective is to design a basic buck (step-down) converter capable of converting a higher DC input voltage (24V) into a regulated lower output voltage (12 V) suitable for sensitive electronic applications. This involves careful selection and design of components such as:

- Inductor (1 mH)
- Capacitor (470 μ F)
- Switching device (MOSFET)
- PWM signal generation
- Load resistance (10 Ω)

The converter will be modelled and simulated in MATLAB Simulink, allowing observation of current, voltage, and switching behaviour in both open-loop and closed-loop modes.

2. To Implement a PI Controller for Output Voltage Regulation

The second objective focuses on implementing a PI controller as the core feedback mechanism to regulate the converter's output voltage. The PI controller is responsible for:

- Continuously comparing the output voltage with a set reference value.
- Minimizing the steady-state error through proportional and integral actions.
- Dynamically adjusting the duty cycle of the PWM signal to maintain voltage stability.

This implementation enhances the converter's adaptability to changes in input voltage and load, which is critical for practical applications.

3. To Analyse the Performance Improvement Using Feedback Control

The final objective is to thoroughly analyse the improvement in performance achieved by adding feedback control. This involves:

- Observing output voltage stability, transient response, and load regulation
- Evaluating how the PI controller minimizes output deviations and maintains a consistent voltage under dynamic conditions

Through waveform analysis and simulation data, the project aims to clearly demonstrate the effectiveness of the PI-controlled feedback system in enhancing the reliability and efficiency of the buck converter.

SCOPE OF RESEARCH

1. DSP controlled Buck Converter

This project can be effectively implemented using a Digital Signal Processing (DSP) controller that operates based on Proportional-Integral (PI) control logic. The DSP controller ensures precise and efficient performance by continuously adjusting system parameters through the PI algorithm.

2. No Hardware Implementation

The project is restricted to MATLAB Simulink-based simulation only. Practical issues like PCB layout, thermal losses, and hardware interfacing are not addressed.

3. Fixed Load Resistance:

The simulation uses a fixed resistive load ($10\ \Omega$), and does not explore variations such as dynamic or nonlinear loads, which are common in real applications.

4. Ideal Switching Assumption:

The simulation assumes ideal switch characteristics (zero ON resistance, infinite OFF resistance, no switching delay), which may not reflect real-world device behaviour.

5. No Input Voltage Ripple:
The input voltage is considered purely DC and ripple-free, whereas actual power sources may exhibit fluctuations or noise.
6. Simplified Control Strategy:
Only a PI controller is implemented. More advanced control strategies like PID, Fuzzy Logic, or Adaptive control are not considered, even though they may offer improved performance.
7. Tuning Not Optimized for All Conditions:
PI controller tuning is done manually or through basic methods and may not be optimal under all load or input conditions.
8. Temperature Effects Ignored:
The influence of temperature on component behaviour (like MOSFET efficiency, capacitor ESR, etc.) is not taken into account.

LITERATURE REVIEW

Erickson and Maksimovic (2001) provided foundational knowledge of buck converter operation, modelling, and control. They emphasized the need for precise feedback mechanisms in variable load conditions.

Krein et al. (1998) investigated the dynamic behavior of DC-DC converters and demonstrated how control methods like PI can stabilize converter output. Research by Texas Instruments and other semiconductor companies highlights the benefits of DSP-based control in power electronic systems, emphasizing speed, flexibility, and digital programmability.

More recent works have demonstrated simulations of closed-loop buck converters using MATLAB/Simulink, validating the effectiveness of PI control in various scenarios (e.g., Zhang et al., 2020). These studies used simulation platforms to tune controller parameters for desired transient and steady-state performance.

Few studies have combined PI control with DSP-based implementation in low-voltage buck converters, especially using MATLAB simulations. Controller parameters are often tuned by trial and error, with little focus on automated tuning based on system behavior.

METHODOLOGY

A buck converter is a type of DC-DC converter used to step down a higher input voltage to a lower output voltage efficiently. The open-loop control is primarily useful for understanding the basic working of a buck converter, analysing the switching behaviour, and observing how the output voltage depends on the duty cycle

- **OPEN LOOP BUCK CONVERTER**

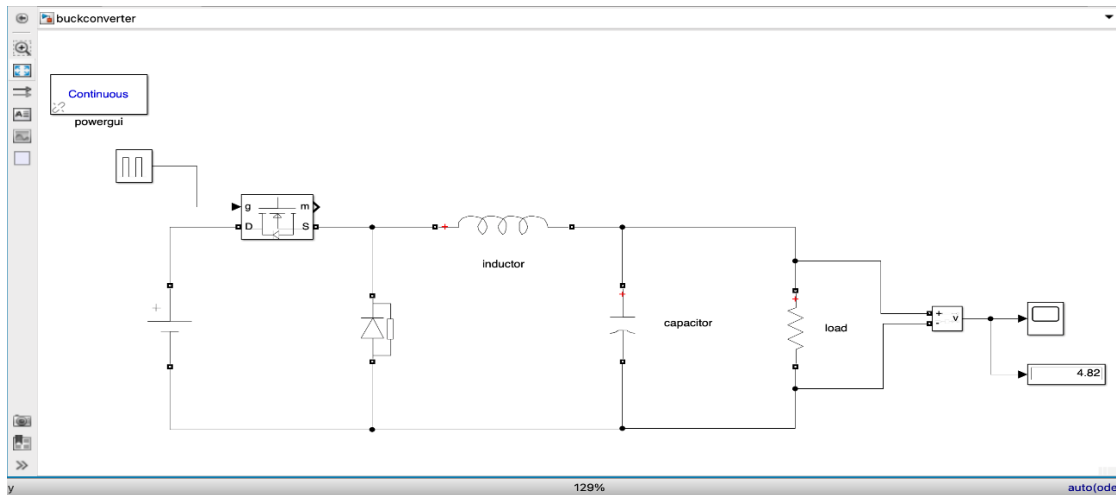


Fig.1

1. Mode 1 (Switch is closed):

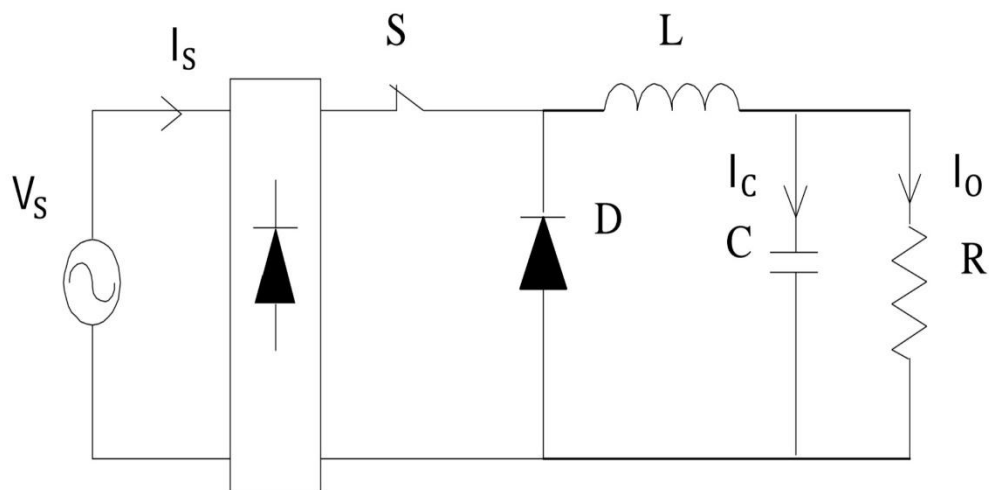


Fig.2

When the controlled switch (e.g. MOSFET) 'S' is on by pulse width modulation (PWM), then input voltage appears across the inductor L, filter capacitor, C and load resistor R and current in inductor L increases linearly. In

the same cycle the capacitor C is charged. During mode 1, the diode reversed biased and resulted from flowing current, the input provides energy to the load as well as to the inductor.

2. Mode 2 (Switch is open):

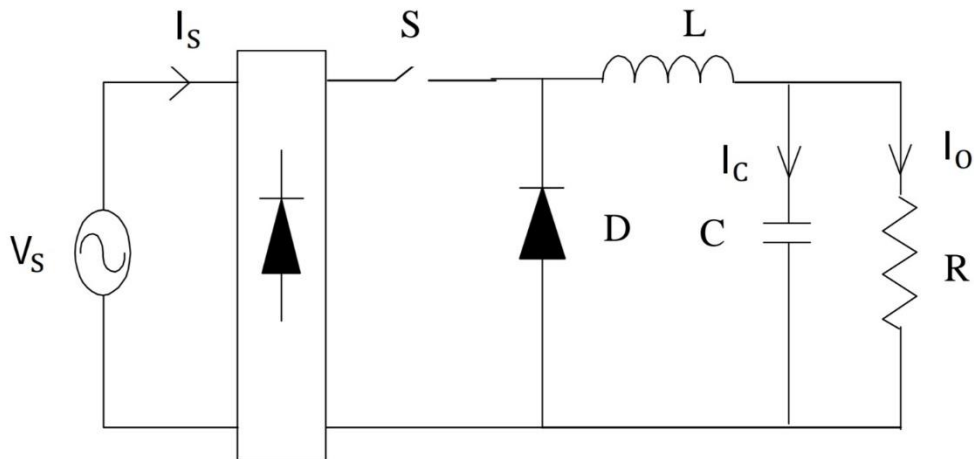


Fig.3

When the controlled switch that is MOSFET is switch off. The voltage across the inductor L is reversed. However, current in the inductor L cannot change instantaneously and the current starts decreasing linearly through inductor L capacitor C, load R and diode D. In this cycle the capacitor is also charged with the energy stored in the inductor.

Parameters:

$$V_o = D \cdot V_{in}$$

$$L = V_o(1-D)/(\Delta I_L f_s)$$

$$C_o = (1-D)/(8L f_s^2) (\Delta V_{co} / V_o)$$

Where

f_s = switching frequency

ΔI_L = Peak to Peak ripple current I_L (assuming 10% of I_L)

ΔV_{co} = voltage ripple (assuming 5% of V_o)

D = Duty cycle.

- The calculated value of Buck converter:

Input voltage (V_{in}) = 12 volts

Output voltage (V_o) = 5 volts

Duty cycle (D) = 0.4

Switching frequency (f_s) = 25 kHz

Resistance(R) = 10 Ω

Inductor (L) = 100 μH

Capacitor (C) = 1000 μF

- **Result waveform graph**

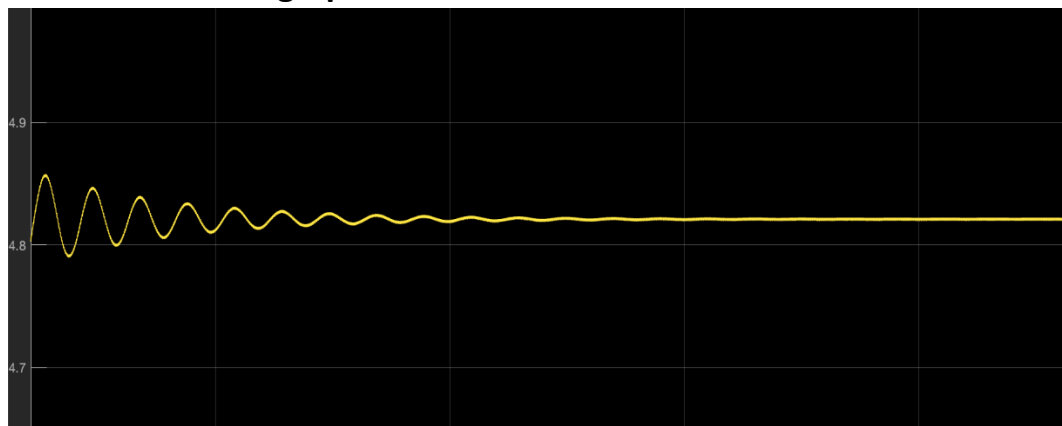


Fig.4

The results of open loop buck converter is shown in figure ,which depicts peak to peak ripple voltage (ΔV_o) is 8.6 Volt and maximum overshoot of 17.3%. Since the design equations assume constant input voltage and constant load under steady state conditions, the variation of input voltage shall result in fluctuation in output Therefore, a closed loop controller is required with optimized parameters to suit the constant voltage output as per requirement of load.

- **CLOSED LOOP BUCK CONVERTER**

Generally, DC-DC converters are the simple power electronic devices that switches one level of input voltage to another level of output voltage. Normally, DC-DC converter output voltage can be controlled using PWM switching with PID controllers in closed loop.

1. System Modelling and Design Parameters

- The buck converter is modelled with key components: inductor (L), capacitor (C), a switch (MOSFET or IGBT), and a diode.
- Operates in Continuous Conduction Mode (CCM) to ensure a steady output current.
- Design parameters are chosen based on desired output voltage, ripple, transient response, and efficiency.

2. PI Controller Design

- A Proportional-Integral (PI) controller is used to regulate the output voltage.
- It reduces the error between the reference voltage and the actual output by adjusting the PWM duty cycle.
- Kp and Ki are tuned to ensure minimal overshoot, fast response, and zero steady-state error.

3. Simulation and Validation

- The system is simulated in MATLAB Simulink.
- Performance is analysed under different input and load conditions.
- Key indicators like voltage regulation, response time, and stability are evaluated.

The blocks used to design the circuit are taken from the library

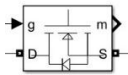
The following blocks are used :

1. Dc voltage source



Use: Provides the input high DC (e.g., 24V) that needs to be stepped down. voltage

2. MOSFET



Acts as the main switching device for the buck converter .
Turns OFF/ON based on PWM signal

3. Inductor , Capacitor , Resistance blocks

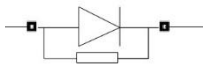


Inductor stores energy when the switch is ON and releases it when OFF .

Capacitor filters the output voltage to ensure a smooth DC voltage output with minimal ripple

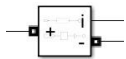
Load resistance represents the output load . The voltage across the resistor is the output voltage we aim to regulate.

4. Diode



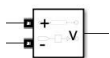
Allows current to flow during the OFF period of the IGBT

5. Current measurement block



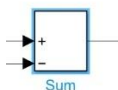
To measure the current passing through the inductor

6. Voltage measurement block



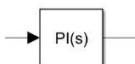
To measure the voltage across the load resistance

7. Sum block



The Sum block's main use is to compare desired vs actual output and generate the error that guides the PI controller for regulating the system

8. PID Controller



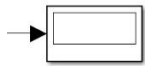
A PID controller improves response and reduces overshoot compared to PI by adding a derivative action.

9. PWM Generator



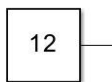
Processes the error signal and generates a control signal to adjust the PWM duty cycle to regulate output voltage

10. Display block



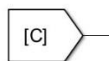
The Display block shows real-time values like output voltage, error, and duty cycle during simulation, helping you quickly monitor and verify the converter's performance.

11. Constant block



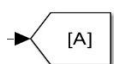
the Constant block is used to provide a fixed reference voltage (e.g., 12V) to the PI controller. This sets the desired output voltage that the system tries to maintain through feedback control.

12. From Block



The From block in Simulink is used to pass signals between distant parts of a model without using direct connecting lines. In your buck converter project, it helps keep the diagram clean and organized by sending signals like output voltage or control signals to other blocks without cluttering the layout.

. GOTO Block



The Goto block sends signals to a From block, letting you transfer data across the model without direct connections, keeping it neat and manageable.

14. Scope



Displays waveforms such as output voltage, current, or control signals for analysis

The following is the circuit of PI controlled closed loop Buck Converter -

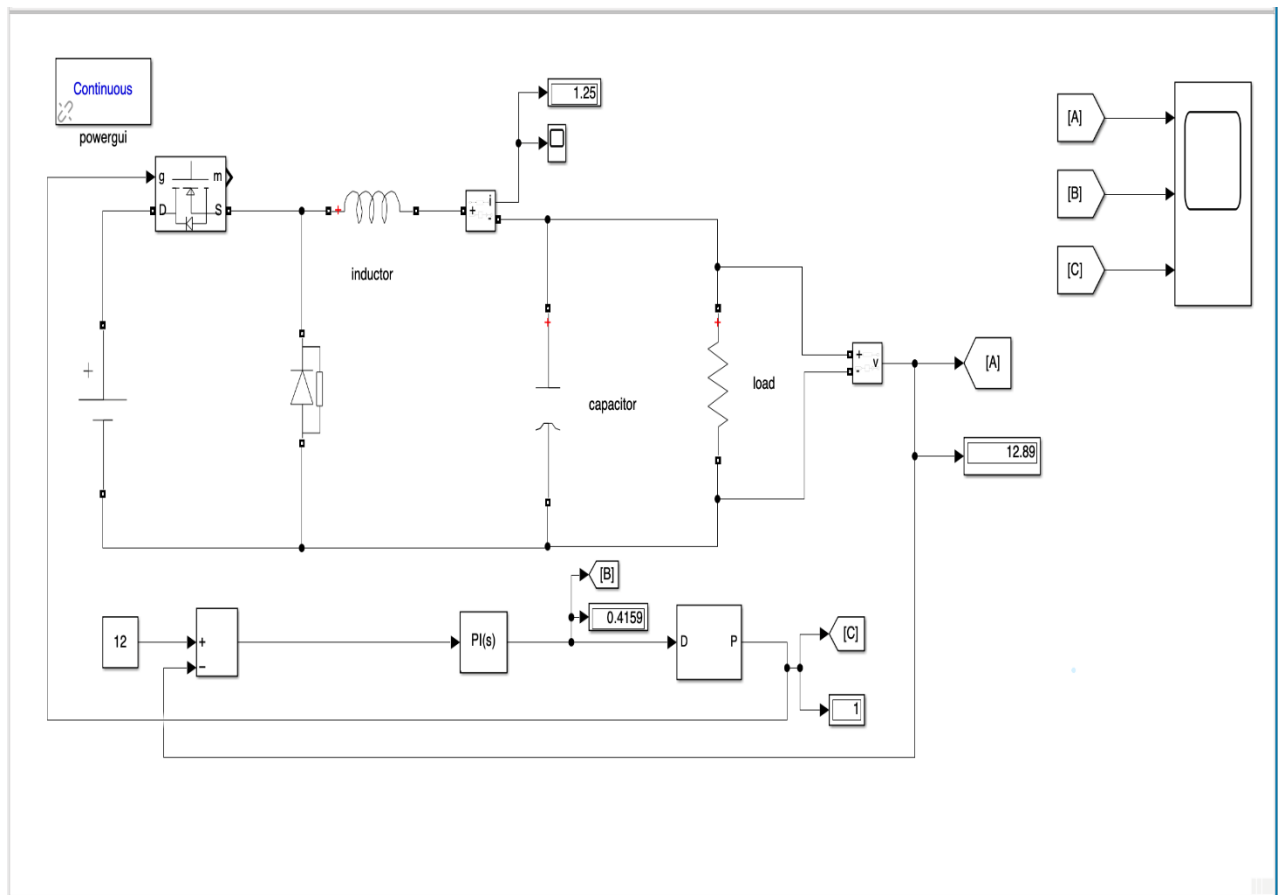


Fig 5

RESULT AND ANALYSIS

PARAMETERS

Vin	24 V
Vout	12 V
Inductance(L)	1 mH
Capacitance(C)	470 μ F
Resistance(R)	10 Ω
Frequency	25 kHz
Duty cycle	0.4
Kp	0.1
Ki	50

- Current v/s Time

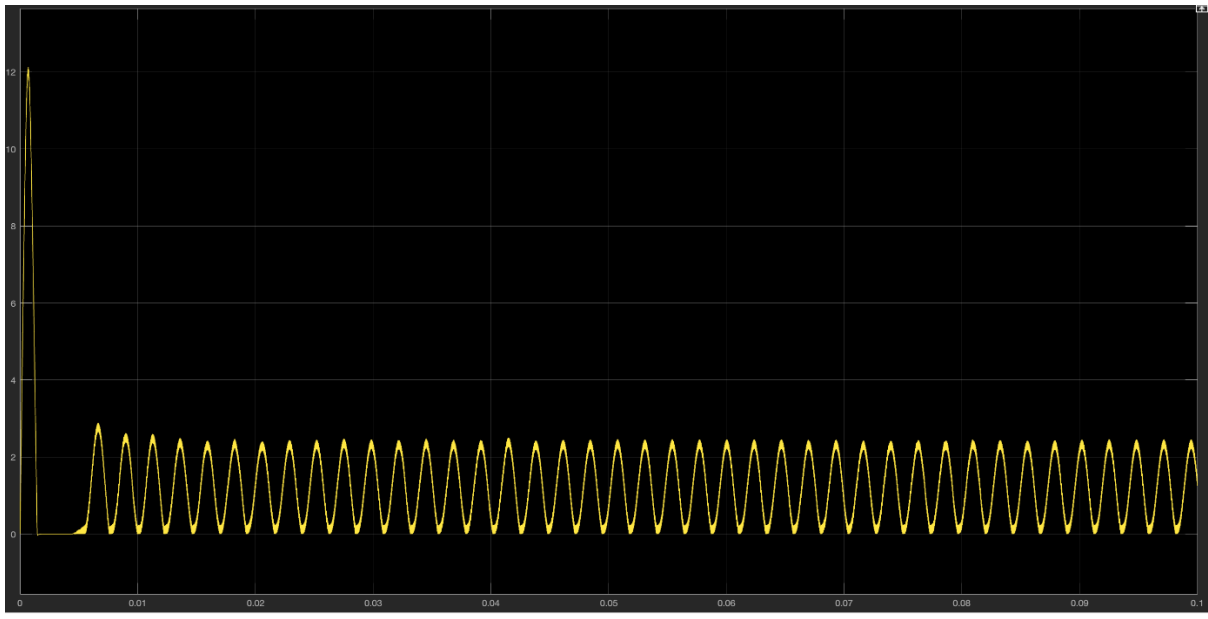


Fig 6

This plot illustrates the inductor current response over time in the PI-controlled buck converter. The waveform shows how the current evolves as the converter reaches a steady state.

At $t = 0$, the current starts from zero and increases as the PI controller ramps up the duty cycle to regulate the output voltage.

Once the converter stabilizes, I_L remains around 1.25 A, indicating that the converter is supplying the load ($10\ \Omega$) with a constant current

$$I = V_{out}/R = 12.89/10 = 1.29\text{ A}$$

- **Vout v/s Time**

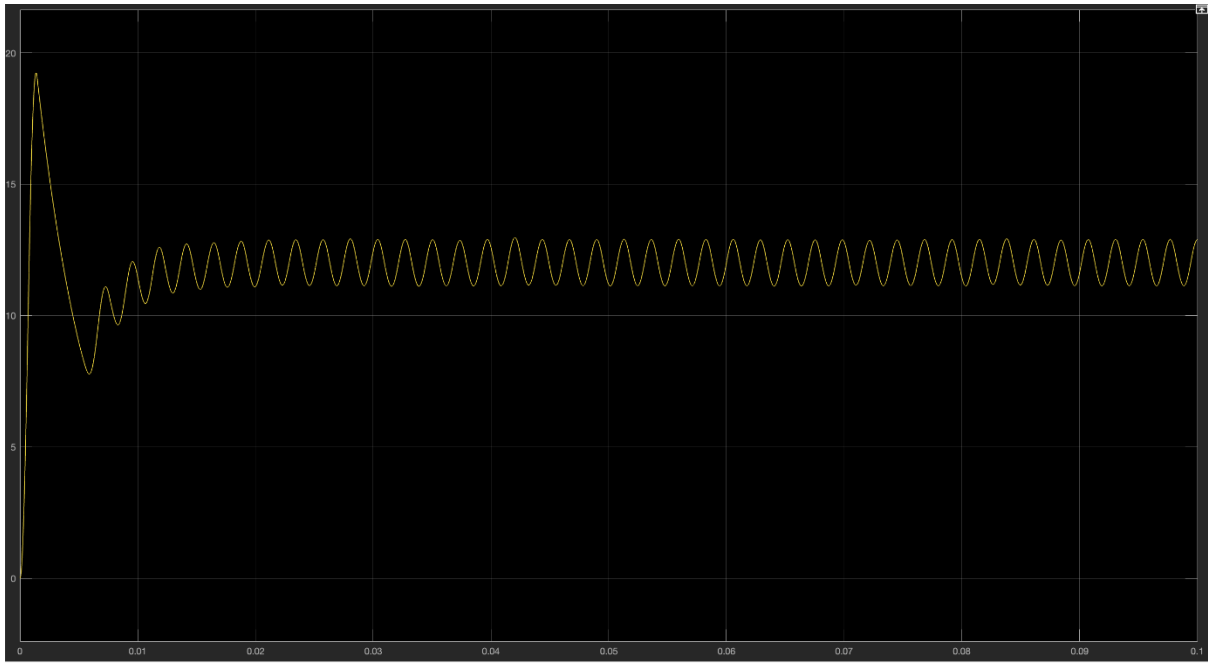


fig 7

This plot shows the output voltage of the buck converter over time under PI-controlled closed-loop operation. The reference voltage (V_{ref}) is set to 12 V.

At time = 0, V_{out} starts from 0 V. It begins to rise quickly as the PI controller adjusts the duty cycle.

The final output voltage stabilizes at 12.89 V, which is a small steady-state error of ~7.4%. The PI controller maintains a nearly constant output despite initial transients.

$$\text{Overshoot} = (V_{out} - V_{ref})/V_{ref} * 100 \%$$

$$= (12.89 - 12)/12 * 100$$

$$= 7.42\%$$

- **Duty Cycle v/s Time**

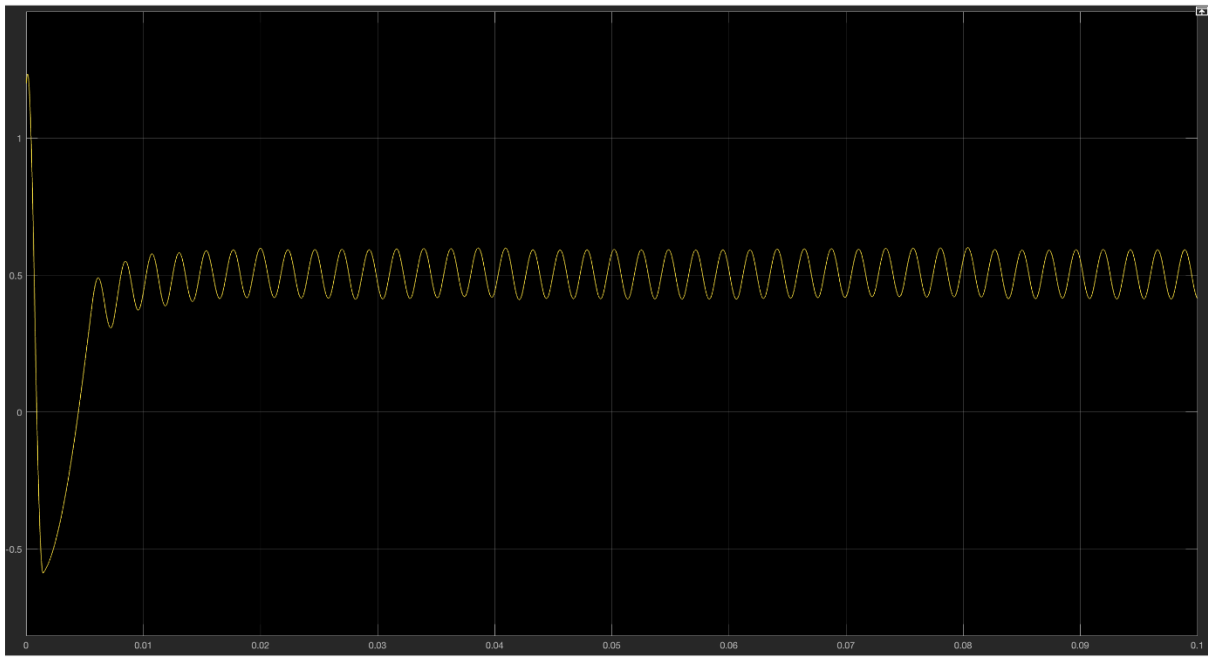


fig 8

This plot shows how the duty cycle is dynamically adjusted by the PI controller over time to regulate the output voltage of the buck converter.

At startup, the PI controller sets an initial duty cycle of 0.4, calculated theoretically for converting 24 V to 12 V ($D = V_{out}/V_{in} = 12/24 = 0.5$; but accounting for losses and non-idealities, an initial lower D is set).

The duty cycle gradually increases from 0.4 and settles around 0.4159, which is the required value to maintain the output at 12.89 V.

The final value of the duty cycle remains stable, indicating successful tuning of the PI controller gains ($K_p = 0.1$, $K_i = 50$).

Variations in Vout , Duty cycle , Overshoot % ,Steady state error by using different Kp and Ki values

Case No.	Kp	Ki	Vout (V)	Duty cycle	Overshoot(%)	Steady state error(%)
1	0.05	20	11.6	0.4030	2.5	3.33
2	0.1	50	12.89	0.4159	7.4	7.42
3	0.2	60	13.1	0.4220	9.1	9.17
4	0.2	100	13.8	0.4300	15.0	15
5	0.3	150	Unstable	>0.45	-	System oscillates

- **ANALYSIS**

Low Kp and Ki (Case 1): Slow response, small steady-state error, very stable.

Moderate Kp and Ki (Case 2): Faster response, small overshoot, good regulation.

High Ki or Kp (Cases 3-4): Faster settling but higher overshoot and risk of instability.

Very high Kp and Ki (Case 5): System becomes unstable—oscillations or divergence in output.

APPLICATIONS

Applications of PI Controlled Closed Loop Buck Converter

A buck converter integrated with a PI controller in closed-loop configuration offers precise output voltage regulation, making it highly suitable for a wide range of real-world applications. Below are key areas where such systems are effectively utilized:

1. Mobile and Portable Device Chargers

Closed-loop buck converters with PI controllers are commonly used in mobile phone chargers, power banks, and other battery-powered devices to ensure consistent voltage supply, protect sensitive circuits, and extend battery life.

2. Renewable Energy Systems

In solar photovoltaic (PV) applications, buck converters are used to step down variable solar panel output voltages to usable levels for battery charging or grid-tied inverters. The PI controller ensures stable voltage despite changes in sunlight intensity or load.

3. Electric Vehicles (EVs)

DC-DC converters with closed-loop control are critical in electric vehicles to supply stable voltage to various low-voltage systems such as lighting, instrumentation, and infotainment from high-voltage battery packs.

4. Industrial Power Supplies

In automated industries, buck converters regulated by PI controllers are used in control panels, PLCs, and sensor power supplies to ensure accurate and stable voltage delivery under varying load conditions.

5. Computer Power Supplies

Modern computer power supplies use multiple DC-DC converters to generate different voltage rails. PI-controlled buck converters ensure that voltage levels for CPU, RAM, and GPU remain consistent even during dynamic load changes.

6. Automotive Electronics

Modern vehicles contain many sensitive electronic modules (like infotainment systems, sensors, and control units) that require precisely regulated voltages.

FUTURE SCOPE OF THE PROJECT

This project presents the successful simulation of a PI-controlled closed-loop buck converter in MATLAB Simulink. While the system performs effectively in regulating output voltage, several areas remain open for further research and enhancement. The future scope includes:

1. Hardware Implementation

The current work is simulation-based. Implementing the buck converter and PI controller on real hardware using microcontrollers or DSPs (such as Arduino, STM32, or TI C2000 series) can validate the simulated results in real-time conditions. Testing on physical circuits would also allow for analysis of thermal behaviour, switching losses, and electromagnetic interference (EMI).

2. Advanced Control Techniques

Integration of advanced control methods such as PID, Fuzzy Logic, Sliding Mode Control, or Model Predictive Control (MPC) can further improve dynamic performance, especially under fast load changes.

Adaptive and self-tuning controllers could be explored to handle varying operating conditions without manual tuning.

3. Efficiency Optimization

Future work can include analysis of converter efficiency by considering non-idealities such as switch losses, inductor core losses, and ESR in capacitors.

Implementation of synchronous rectification can improve efficiency at low voltage levels.

4. Renewable Energy Integration

This converter model can be extended to operate with solar panels or energy storage systems where input voltage fluctuates frequently. Real-time MPPT (Maximum Power Point Tracking) can be added along with the PI controller.

5. Digital Control Implementation

The PI control algorithm can be implemented using digital signal processing techniques in embedded platforms to improve precision, speed, and

programmability. Integration with IoT systems for remote monitoring and adaptive control is also a growing area.

CONCLUSION -

Step down switching regulators are the backbone of electronic equipments. The designing of buck converters has been carried out for constant voltage applications considering KP and KI are the performance parameter for PI controller. Buck converter has been designed to deliver 48 volts DC to a 100 watt load. Performance and applicability of this converter is presented on the basis of simulation in MATLAB SIMULINK. Buck converters are employed for low power applications below 150 W and with voltages below 230V. The parametric variation analysis of buck converter have been carried out for constant voltage applications considering. The design concepts are validated through simulation and results obtained show that a closed loop system using buck converter will be highly stable with high efficiency. Better efficiency due to: moderate duty cycles, lower voltage MOSFETs and rectifiers, and reduced switching losses due to reduced peak-to-peak voltage swing.

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